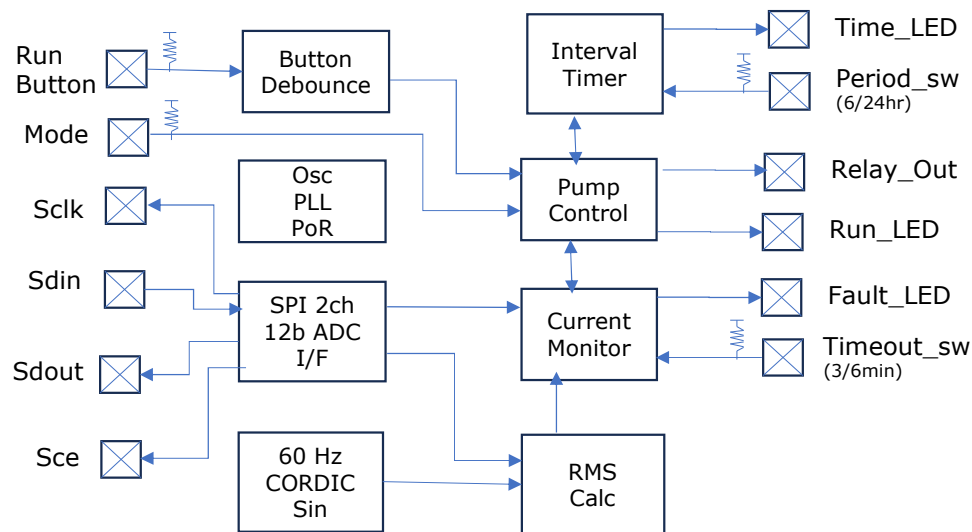


LPC Block Diagram



Typical Applications

- Automatic water-level control for residential pump systems Reliable float-switch replacement using motor current.
- RMS current based pump control in compact commercial equipment Solid-state activation and cutout for small tanks and reservoirs.
- Embedded pump logic for OEM assemblies Deterministic current-profile switching without mechanical floats.

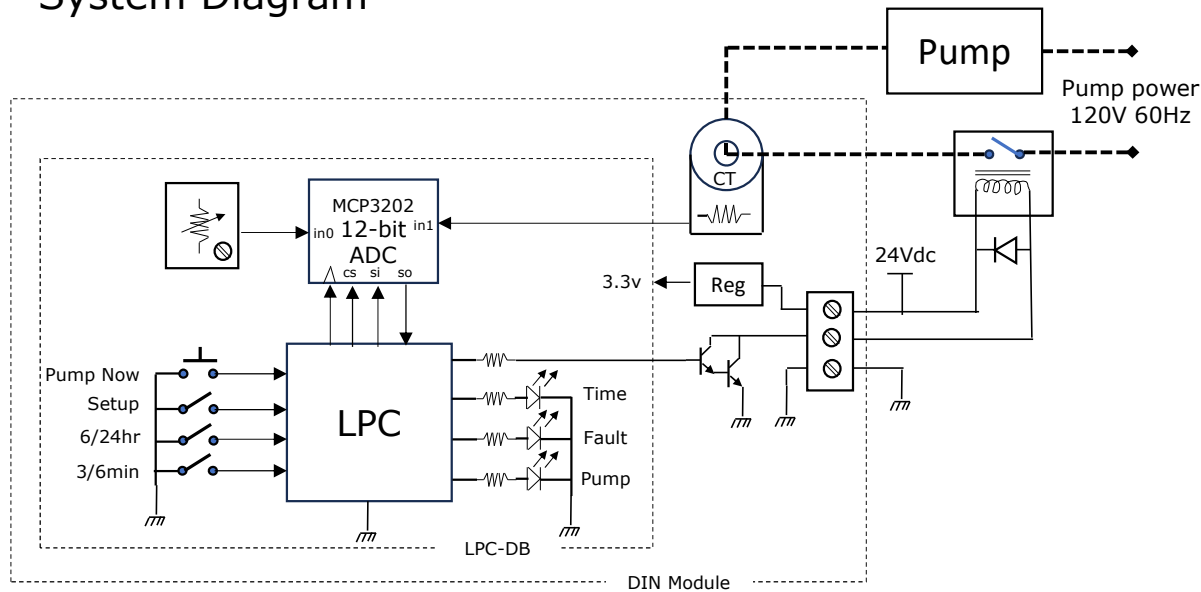
Purpose

- Deterministic current-sensing controller that replaces mechanical float switches with reliable automatic pumping.

Features

- Single CT input with 60 Hz CORDIC RMS engine for stable motor-current sensing
- 12-bit external SPI ADC for precise 15Khz current-profile capture
- Front-panel threshold trim sets cutout margin below run current
- Current-cutout logic detects empty-tank, jam, or timeout
- Automatic periodic pumping (6 hr / 24 hr DIP-selectable cycle)
- Timed pump-out window (3 min / 6 min DIP-selectable max run)
- Manual pump-out button for immediate emptying
- Manual mode follows button directly for setup and testing
- Status LEDs:
 - RUN (on-pump),
 - FAULT (too long/too short/overcurrent),
 - TIME (pending cycle)
- TIP122 open collector relay drive transistor
- Three terminal system interface (Pwr, Gnd, OC relay drive)
- Integrated PLL, oscillator, and Power on Reset
- Always-works architecture. No firmware, no crashes, no jitter
- DIN-module integration with upper logic/HMI board and lower power/relay board
- Development board (DB) with emulator header

System Diagram



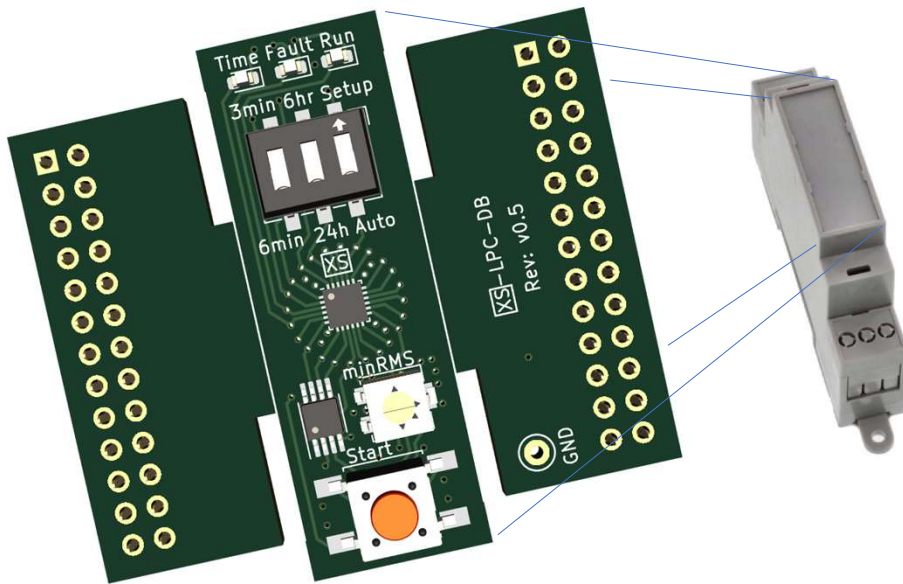
Description

The LPC controls a pump using CT-based current sensing and timed operation. At power-up, manual button press, or after the selected 6 hr / 24 hr interval, the controller initiates a pump-out cycle. The relay output drives the pump, and the CT, is digitized at 15KHz for monitor of motor current to detect the drop that indicates an empty tank. During setup a trimpot is used with hi/lo feedback to set the current limit. The LPC terminates a pump-out cycle on current-cutout, timeout (3 min / 6 min), or fault conditions including over-current and short-cycle behavior. A breathing Time LED indicates progress toward the next scheduled pump-out, while the Pump LED asserts during active relay drive. Fault conditions produce coded blink patterns on the Fault LED. The system operates from a 24 V supply with local 3.3 V regulation and uses an externally protected open-collector driver for the pump relay coil.

System Behavior

Pump cycles run on interval, power-up, or manual request.
Current transformer CT derived 60Hz RMS current determines empty-tank cutoff.
Fault logic enforces safe, deterministic termination.

LPC-DB Development Board



Integration Value

Reference design enables immediate product deployment.
Full bring-up, test, and emulation environment (emulator tabs not shown)
DIN-module form factor supports certification and OEM use.

Purpose

The LPC-DB provides a complete environment for chip bring-up, product development, and system evaluation. All front-panel controls, indicators, and relay interfaces are exposed for direct testing, while the emulator header supports firmware iteration and feature refinement. The board fits directly into a 0.7" DIN module, enabling immediate deployment as a functional pump controller and serving as a certifiable reference design for OEM integration. The LPC-DB allows customization, observation, and trial of system behavior while remaining fully usable as-shipped.

Features

- Complete LPC chip bring-up platform
- Full front-panel control interface
- Relay drive and CT input
- Emulator header 'wings' for feature testing
- Supports firmware and logic iteration
- Direct DIN-module mechanical fit
- Suitable for regulatory certification
- Reference design for OEM products
- Immediate functional pump controller
- Adjustable interval and timeout logic
- Real-time LED status indicators
- Customizable system behavior evaluation